Product Requirements Document (PRD)

E-Ticketing System for Mobile Phone Repair Services

Product Name: E-Ticketing Service HP
Platform: Web Application
Target Users: Mobile phone repair shops, administrators, technicians, and customers
Main Goal: Streamline the repair workflow from device intake to completion while automatically notifying customers through WhatsApp.

1. Background

Many mobile phone repair shops still record repair jobs manually using notebooks, spreadsheets, or WhatsApp chats. This makes ticket tracking, repair monitoring, and customer communication inefficient.

The E-Ticketing System provides a centralized workflow for registering incoming devices, tracking repair progress, recording costs and technician activities, and notifying customers when important status changes occur.

• Device received

• Under inspection

• Waiting for customer approval

• Under repair

• Waiting for spare parts

• Delayed

• Repair completed

• Ready for pickup

• Picked up

2. Problem Statement

• Repair jobs are still recorded manually.

• Staff cannot quickly determine the current status of a customer's device.

• Customers must repeatedly contact the shop to ask for updates.

• Administrators and technicians have difficulty monitoring active workloads.

• There is no clear and auditable history of ticket status changes.

• Customers may not immediately know when a repair is completed or delayed.

• Generating operational reports is difficult.

3. Product Goals

• Create repair tickets quickly.

• Store customer and device information.

• Record customer complaints and device issues.

• Generate unique ticket numbers automatically.

• Manage the complete repair status lifecycle.

• Assign tickets to technicians.

• Record cost estimates and expected completion dates.

• Request customer approval when required.

• Send WhatsApp notifications.

• Provide a public ticket tracking page.

• Store a complete history of ticket activities.

• Provide operational dashboards and reports.

4. User Roles

4.1 Super Admin / Administrator

• Log in and manage users and roles.

• Create and manage repair tickets.

• Register customers and devices.

• View and update ticket information.

• Assign technicians.

• Record payments.

• Resend WhatsApp notifications.

• Complete device handover to customers.

• Access dashboards and reports.

4.2 Technician

• View assigned tickets.

• Review customer complaints and device details.

• Update repair status.

• Add inspection and diagnosis results.

• Add repair cost estimates.

• Add technician notes.

• Mark tickets as waiting for spare parts.

• Mark repairs as completed.

• Upload device condition photos when needed.

4.3 Customer

Customers do not need to create an account for the MVP.

• Receive a ticket number.

• Open a public tracking page.

• View repair status and estimated completion date.

• Receive delay information.

• Receive WhatsApp notifications.

• View payment status.

• Receive confirmation when the device is ready for pickup.

5. Main User Flow

5.1 Device Intake

• The customer brings a device to the repair shop.

• The administrator opens the Create Ticket page.

• Customer information is entered or selected.

• Device information is entered.

• The customer's complaint is recorded.

• The physical condition and included accessories are recorded.

• The system generates a unique ticket number.

• The ticket is saved.

• The customer receives the ticket information through WhatsApp.

Example Ticket Number: SRV-20260819-0001

5.2 Inspection and Diagnosis

• The ticket is assigned to a technician.

• The technician changes the status to CHECKING.

• The technician performs an inspection.

• Diagnosis results are recorded.

• An estimated cost and completion time are entered.

• If approval is required, the status changes to WAITING_APPROVAL.

• The customer receives a notification.

5.3 Repair Process

• After approval, the repair process begins.

• The status changes to REPAIRING.

• If spare parts are required, the status changes to WAITING_PART.

• If the estimated completion time cannot be met, the status changes to DELAYED.

• The customer receives a delay notification.

• After the repair is completed, the ticket is marked COMPLETED or READY_PICKUP.

• The customer receives a notification that the device is ready.

5.4 Device Pickup

• The customer arrives at the shop.

• The administrator searches for the ticket.

• Payment status is verified.

• The device is handed over.

• The ticket status changes to PICKED_UP.

• The ticket is closed.

6. Ticket Statuses

Status	Description
RECEIVED	The device has been received by the repair shop.
CHECKING	The device is currently being inspected.
WAITING_APPROVAL	Waiting for customer approval before continuing.
APPROVED	The customer approved the repair.
REJECTED	The customer rejected the repair.
REPAIRING	The device is currently being repaired.
WAITING_PART	The repair is waiting for spare parts.
DELAYED	The repair process is delayed.
COMPLETED	The repair work has been completed.
READY_PICKUP	The device is ready to be collected.
PICKED_UP	The device has been collected by the customer.
CANCELLED	The repair ticket was cancelled.

Every status change must be stored in the Ticket Status History for auditing and operational monitoring.

7. Functional Requirements

FR-01: Authentication

• Login

• Logout

• User management

• Role-based access control

• Password reset

Initial roles: Super Admin, Administrator/Cashier, and Technician.

FR-02: Customer Management

• Create customers

• View customer list

• Search by name or WhatsApp number

• Edit customer information

• View customer repair history

Customer data: name, WhatsApp number, optional email, and optional address.

FR-03: Device Management

• Store device brand

• Store model

• Store IMEI

• Store color

• Store storage capacity

• Record physical condition

• Record accessories left with the device

Examples: charger, SIM card, case, and memory card.

FR-04: Create Ticket

• Generate ticket number automatically

• Select customer and device

• Record complaint

• Record physical condition

• Assign technician

• Set priority

• Set estimated completion date

• Add notes

• Set initial status

Priority levels: Low, Normal, High, and Urgent.

FR-05: Ticket Management

• View all tickets

• Search tickets

• Filter by status, technician, and date

• View ticket details

• Edit ticket information

• Cancel tickets

• Generate service receipts

FR-06: Technician Assignment

• Assign one or more technicians to a ticket

• View assigned tickets

• View active jobs

• View completed jobs

• View overdue and delayed jobs

FR-07: Diagnosis and Cost Estimate

• Record inspection results

• Record root cause

• Record recommended action

• Record required spare parts

• Record spare part costs

• Record labor costs

• Calculate estimated total

• Set estimated completion time

FR-08: Customer Approval

• Send repair estimate for approval

• Set status to WAITING_APPROVAL

• Record approval result

• Change status to APPROVED or REJECTED

For the MVP, approval may be recorded manually. A future version may provide approval links directly in WhatsApp.

FR-09: WhatsApp Notifications

• Notify when a ticket is created

• Notify when approval is required

• Notify when the repair is delayed

• Notify when the device is ready for pickup

• Store delivery status, time, recipient, message, and errors

• Support retry for failed notifications

8. WhatsApp Notification Templates

Ticket Created

Hello {{customer_name}}, your device has been received for repair.
Ticket Number: {{ticket_number}}
Status: Device Received
Track your repair: {{tracking_url}}

Waiting for Approval

Hello {{customer_name}}, the inspection result for your device is available.
Estimated Cost: Rp{{estimated_cost}}
We are waiting for your approval to continue the repair.

Repair Delayed

Hello {{customer_name}}, your repair is taking longer than expected.
Ticket Number: {{ticket_number}}
Reason: {{delay_reason}}
Updated Estimate: {{new_estimate}}

Repair Completed

Hello {{customer_name}}, your device repair has been completed and is ready for pickup.
Ticket Number: {{ticket_number}}
Total Cost: Rp{{total_cost}}

9. Public Customer Tracking

Customers can access a public tracking page without logging in.

Example URL: /track/SRV-20260819-0001

• Ticket number

• Partially masked customer name for privacy

• Device brand and model

• Customer complaint

• Current status

- Status timeline
- Estimated completion date
- Delay information, when applicable
- Ready-for-pickup information

Sensitive information such as the full IMEI must not be exposed on public tracking pages.

10. Dashboard

- Total tickets today
- New tickets
- Tickets under repair
- Tickets waiting for spare parts
- Delayed tickets
- Completed tickets
- Devices ready for pickup
- Recent tickets
- Tickets approaching their deadline
- Overdue tickets
- Technician performance
- Repair volume by device brand

11. Ticket Detail

- **Customer Information:** name and WhatsApp number
- **Device Information:** brand, model, IMEI, and color
- **Service Information:** ticket number, complaint, technician, priority, status, intake date, and estimated completion date
- **Diagnosis:** inspection result, cause, and recommendation
- **Costs:** labor, spare parts, total, payments, and outstanding balance
- **Timeline:** complete history of ticket status changes

12. Payment

- Unpaid
- Deposit
- Paid in Full

Payment records must include total cost, deposit amount, total amount paid, remaining balance, payment method, payment date, and receiving administrator.

Supported payment methods: Cash, Bank Transfer, QRIS, and E-Wallet.

13. Service Receipt

- Repair shop name
- Ticket number
- Customer name and WhatsApp number

• Device information

• Customer complaint

• Physical condition

• Included accessories

• Estimated completion date

• Service terms and conditions

• QR code for ticket tracking

The receipt should be printable, downloadable as a PDF, and optionally sent through WhatsApp.

14. Non-Functional Requirements

Performance

• The dashboard should load quickly.

• Ticket search should return results within a few seconds.

• The system should support a growing number of tickets without significant performance degradation.

Security

• Passwords must be securely hashed.

• Role and permission checks must be enforced.

• Internal endpoints must require authentication.

• Customer data must be protected.

• Full IMEI values must not be exposed publicly.

• Important changes must be recorded in an activity log.

Reliability

• Status changes must not be lost.

• Failed WhatsApp notifications must be retryable.

• Ticket history must be protected with an audit trail.

15. Core Database Entities

```
users
roles
permissions

customers
devices

tickets
ticket_assignments
ticket_status_histories
ticket_notes

diagnoses
ticket_parts

payments
whatsapp_notifications
activity_logs
```

Main relationship:

```
Customer
   |
   ■■■ Devices
          |
          ■■■ Tickets
                 ■■■ Status Histories
                 ■■■ Technician Assignments
                 ■■■ Diagnoses
                 ■■■ Parts
                 ■■■ Payments
                 ■■■ Notifications
```

16. MVP Scope

■ Authentication

■ Administrator and Technician roles

■ Customer management

■ Device management

■ Create ticket

■ Automatic ticket number generation

■ Ticket list and ticket detail

■ Status management

■ Ticket status history

■ Technician assignment

■ Diagnosis

■ Cost estimation

■ Delay information

■ WhatsApp notifications

■ Public ticket tracking

■ Payment recording

■ Dashboard

■ PDF service receipt

■ Basic reporting

17. Future Development

Customer Approval via WhatsApp

Provide secure approval actions directly from WhatsApp, such as Approve Repair and Reject Repair.

Customer Portal

• Customer login

• Repair history

• Invoice access

• Service ratings and feedback

Spare Part Inventory

• Spare part stock management

• Stock in and stock out

• Low stock notifications

• Automatic spare part usage linked to repair tickets

Multiple Branches

Support multiple repair shop branches using branch-based data ownership. Central administrators can monitor all branches.

Analytics

• Repair revenue

• Most productive technicians

• Most frequently repaired brands

• Most common issues

• Average repair duration

• Number of delayed tickets

AI Integration

• Complaint classification

• Possible issue recommendations

• Repair duration estimation based on historical data

• Customer service chatbot for ticket tracking

18. Recommended Technology Stack

Backend: Laravel 12+, PHP 8.3+, Laravel Queue, Laravel Scheduler

Frontend Option A: Laravel + Inertia + React + TypeScript + Tailwind CSS

Frontend Option B for Faster MVP: Laravel + Livewire + Tailwind CSS

Database: PostgreSQL or MySQL

Queue and Cache: Redis

WhatsApp: Official WhatsApp API provider with API and webhook support

Storage: Local storage for development and S3-compatible object storage for production

Deployment: Docker, Nginx, PHP-FPM, PostgreSQL/MySQL, Redis, and Queue Workers

19. Success Metrics

• A new ticket can be created in less than 3 minutes.

• Administrators can quickly find tickets by ticket number or WhatsApp number.

• All important status changes are recorded.

• Customers can check repair status without contacting the repair shop.

• Repair completion notifications are sent automatically.

• Manual customer inquiries about repair status are reduced.

• Administrators can monitor all active repairs from a single dashboard.

20. Product Summary

The E-Ticketing System for Mobile Phone Repair Services manages the complete repair lifecycle:

```
Device Intake
   ↓
Ticket Created
   ↓
Inspection
   ↓
Cost Estimate
   ↓
Customer Approval
   ↓
Repair
   ↓
Delay / Waiting for Parts
   ↓
Repair Completed
   ↓
WhatsApp Notification
   ↓
Device Pickup
   ↓
Ticket Closed
```

The MVP focuses on structured repair operations, easy status monitoring, complete ticket history, and proactive customer communication.